DSE-3: Data Science – (Unit-1)

What is Data Science? Data science is a deep study of the massive amount of data, which involves extracting meaningful insights from raw, structured, and unstructured data that is processed using the scientific method, different technologies, and algorithms. Data science is all about:

- Asking the correct questions and analyzing the raw data.
- Modelling the data using various complex and efficient algorithms.
- Visualizing the data to get a better perspective.
- Understanding the data to make better decisions and finding the final result.

Need for Data Science: Following are some main reasons for using data science technology:

- With the help of data science technology, we can convert the massive amount of raw and unstructured data into meaningful insights.
- Data science technology is opting by various companies, whether it is a big brand or a start-up. Google, Amazon, Netflix, etc, which handle the huge amount of data, are using data science algorithms for better customer experience.
- Data science is working for automating transportation such as creating a self-driving car, which is the future of transportation.
- Data science can help in different predictions such as various survey, elections, flight ticket confirmation, etc.

Types of Data Science Job: If you learn data science, then you get the opportunity to find the various exciting job roles in this domain. Some of the job roles are given below:

1. Data Analyst
2. Machine learning expert
3. Data engineer
4. Data Scientist

1. Data Analyst: Data analyst is an individual, who performs mining of huge amount of data, models the data, looks for patterns, relationship, trends, and so on. At the end of the day, he comes up with visualization and reporting for analyzing the data for decision making and problem-solving process.

Skill required: For becoming a data analyst, you must get a good background in **mathematics, business intelligence, data mining**, and basic knowledge of **statistics**. You should also be familiar with some computer languages and tools such as **MATLAB, Python, SQL, Hive, Pig, Excel, SAS, R, JS, Spark**, etc.

2. Machine Learning Expert: The machine learning expert is the one who works with various machine learning algorithms used in data science such as **regression, clustering, classification, decision tree, random forest**, etc.

Skill Required: Computer programming languages such as Python, C++, R, Java, and Hadoop. You should also have an understanding of various algorithms, problem-solving analytical skill, probability, and statistics.

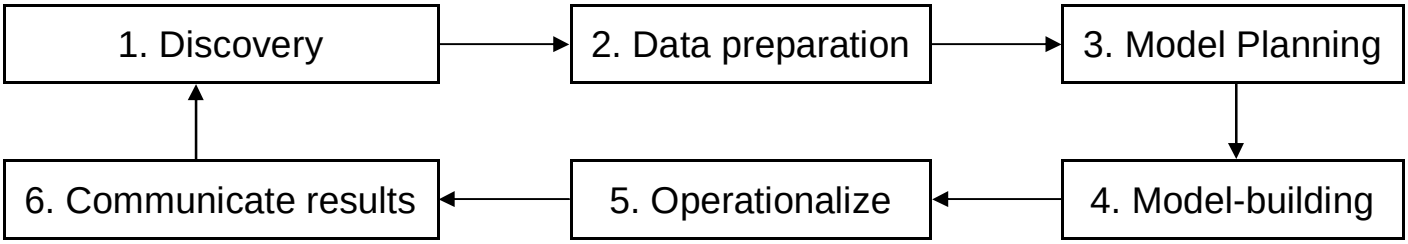
3. Data Engineer: A data engineer works with massive amount of data and responsible for building and maintaining the data architecture of a data science project. Data engineer also works for the creation of data set processes used in modeling, mining, acquisition, and verification.

Skill required: Data engineer must have depth knowledge of **SQL, MongoDB, Cassandra, HBase, Apache Spark, Hive, MapReduce**, with language knowledge of **Python, C/C++, Java, Perl**, etc.

4. Data Scientist: A data scientist is a professional who works with an enormous amount of data to come up with compelling business insights through the deployment of various tools, techniques, methodologies, algorithms, etc.

Skill required: To become a data scientist, one should have technical language skills such as **R, SAS, SQL, Python, Hive, Pig, Apache spark, MATLAB**. Data scientists must have an understanding of Statistics, Mathematics, visualization, and communication skills.

Data Science Lifecycle: The life-cycle of data science is explained as below diagram.



The main phases of data science life cycle are given below:

1. Discovery: The first phase is discovery, which involves asking the right questions. When any data science project is started, the first step is to determine what are the basic requirements, priorities, and project budget. It is also required to determine all the requirements of the project such as the number of people, technology, time, data, an end goal, and then the business problem on first hypothesis level can be framed.

2. Data preparation: Data preparation is also known as Data Munging. In this phase, we need to perform the following tasks:

- Data cleaning
- Data Reduction
- Data integration
- Data transformation

After performing all the above tasks, we can easily use this data for our further processes.

3. Model Planning: In this phase, the various methods and techniques to establish the relation between input variables are determined. We will apply exploratory data analytics (EDA) by using various statistical formula and visualization tools to understand the relations between variable and to see what data can inform us. Common tools used for model planning are:

- SQL Analysis Services
- R
- SAS
- Python

4. Model-building: In this phase, the process of model building starts. Datasets for training and testing purpose are created and then different techniques such as association, classification, and clustering are applied to build the model. Following are some common Model building tools:

- SAS Enterprise Miner
- WEKA
- SPCS Modeller
- MATLAB

5. Operationalize: In this phase, the final reports of the project, along with briefings, code, and technical documents is delivered. This phase provides a clear overview of complete project performance and other components on a small scale before the full deployment.

6. Communicate results: In this phase, it is checked to see if the goal is reached, which was set on the initial phase and then the findings and final result is communicated with the business team.

Data Scientist's Tool Box: Following are some tools required for data science:

- **Data Analysis tools:** R, Python, Statistics, SAS, Jupyter, R Studio, MATLAB, Excel, RapidMiner.
- **Data Warehousing:** ETL, SQL, Hadoop, Informatica/Talend, AWS Redshift
- **Data Visualization tools:** R, Jupyter, Tableau, Cognos.
- **Machine learning tools:** Spark, Mahout, Azure ML studio.

Turning data into actionable knowledge: To produce actionable insights we require clarity on our business and its products, our value proposition, the content on our website, and our plan for marketing Data analysts and scientists are necessary to identify crucial information and build insights for our organization. The list below shows ways to turn big data into actionable insights:

1. Data Visualization: It is necessary not only to have data but also to analyze and synthesize it. Synthesis of data enables us to find patterns. After figuring out patterns, it is necessary to bring data into visualization. Data visualizations help analyze more the relevant information amidst variables, capture and communicate patterns, and build predictive models. Data visualization is an important step to identify patterns and trends, which are otherwise ignored. Data visualization helps to find insights more accurately when compared to tabular data.

2. Pattern Recognition: Pattern recognition is one of the important steps in turning data into actionable insights, which helps to go from information to knowledge. It is important to understand that not all the patterns are relevant. The potential implications of each pattern must be reviewed accurately and must answer all questions about the patterns.

3. Insight Articulation: Insight articulation is articulating the insight that emerged out of each pattern. This strategy helps capture insights and place it on a flip chart, which enables us to track each insight.

4. Insight Incubation: This strategy is a crucial one, which states that we and our team must be given some time away from the board. This helps us to take a fresh look at identifying the insights and states the need for changes if any.

5. Reaction Assimilation: After determining insights, it is necessary to gather the reactions of all the concerned workers who were involved during the initial stages. This helps to decide whether the insights resonate and compel the user enough to make key decisions.

Applications of Data Science:

- **Image recognition and speech recognition:** Data science is currently used for Image and speech recognition. When we upload an image on Face book and start getting the suggestion to tag to our friends. This automatic tagging suggestion uses image recognition algorithm, which is part of data science. When we say something using, "Ok Google, Siri, Cortana", etc., and these devices respond as per voice control. This is possible with speech recognition algorithm.
- **Gaming world:** In the gaming world, the use of Machine learning algorithms is increasing day by day. EA Sports, Sony, Nintendo are widely using data science for enhancing user experience.
- **Internet search:** When we want to search for something on the internet, then we use different types of search engines such as Google, Yahoo, Bing, Ask, etc. All these search engines use the data science technology to make the search experience better, and we can get a search result with a fraction of seconds.
- **Transport:** Transport industries are also using data science technology to create self-driving cars. With self-driving cars, it will be easy to reduce the number of road accidents.
- **Healthcare:** In the healthcare sector, data science is providing lots of benefits. Data science is being used for tumour detection, drug discovery, medical image analysis, virtual medical bots, etc.
- **Recommendation systems:** Most of the companies, such as Amazon, Netflix, Google Play, etc. are using data science technology for making a better user experience with personalized recommendations. Such as, when we search for something on Amazon, we get suggestions for similar products. This is because of data science technology.
- **Risk detection:** Finance industries always had an issue of fraud and risk of loss, but with the help of data science, this can be rescued. Most of the

finance companies are looking for the data scientist to avoid risk and any type of losses with an increase in customer satisfaction.

Introduction to the tools that will be used in building data analysis software:

Git: Git is an open-source distributed version control system. It is designed to handle minor to major projects with high speed and efficiency. It is developed to co-ordinate the work among the developers. The version control allows us to track and work together with our team members at the same workspace. Git is foundation of many services like **GitHub** and **GitLab,** but we can use Git without using any other Git services. Git can be used **privately** and **publicly**. Git was created by **Linus Torvalds** in **2005** to develop Linux Kernel. It is also used as an important distributed version-control tool for **the DevOps**. Git is easy to learn, and has fast performance. It is superior to other SCM tools like Subversion, CVS, Perforce, and ClearCase.

Features of Git: Some remarkable features of Git are as follows:

- **Open Source:** Git is an **open-source tool**. It is released under the **GPL** (General Public License) license.
- **Scalable**: Git is **scalable**, which means when the number of users increases, the Git can easily handle such situations.
- **Distributed:** One of Git's great features is that it is distributed. Distributed means that instead of switching the project to another machine, we can create a "clone" of the entire repository. Also, instead of just having one central repository, every user has its own repository that contains the entire commit history of the project. We do not need to connect to the remote repository; the change is just stored on our local repository. If necessary, we can push these changes to a remote repository.
- **Security:** Git is secure. It uses the SHA1 (Secure Hash Function) to name and identify objects within its repository. Files and commits are checked and retrieved by its checksum at the time of checkout. It stores its history in such a way that the ID of particular commits depends upon the complete development history leading up to that commit. Once it is published, one cannot make changes to its old version.
- **Speed:** Git is very fast, so it can complete all the tasks in a while. Most of the git operations are done on the local repository, so it provides a huge speed. Also, a centralized version control system continually communicates with a server somewhere. Performance tests conducted by Mozilla showed that it was extremely fast compared to other VCSs. Fetching version history from a locally stored repository is much faster than fetching it from the remote server. The core part of Git is written in C, which ignores runtime overheads associated with other high-level languages. Git was developed to work on the Linux kernel; therefore, it is capable enough to handle large repositories effectively. From the beginning, speed and performance have been Git's primary goals.

- **Supports non-linear development**: Git supports seamless branching and merging, which helps in visualizing and navigating a non-linear development. A branch in Git represents a single commit. We can construct the full branch structure with the help of its parental commit.
- **Branching and Merging**: Branching and merging are the great features of Git, which makes it different from the other SCM tools. Git allows the creation of multiple branches without affecting each other. We can perform tasks like creation, deletion, and merging on branches, and these tasks take a few seconds only. Below are some features that can be achieved by branching:
    - We can create a separate branch for a new module of the project, commit and delete it whenever we want.
    - We can have a production branch, which always has what goes into production and can be merged for testing in the test branch.
    - We can create a demo branch for the experiment and check if it is working. We can also remove it if needed.
    - The core benefit of branching is if we want to push something to a remote repository, we do not have to push all of our branches. We can select a few of our branches, or all of them together.
- **Data Assurance**: The Git data model ensures the cryptographic integrity of every unit of our project. It provides a unique commit ID to every commit through a SHA algorithm. We can retrieve and update the commit by commit ID. Most of the centralized version control systems do not provide such integrity by default.
- **Staging Area**: The Staging **area** is also a unique functionality of Git. It can be considered as a preview of our next commit, moreover, an intermediate area where commits can be formatted and reviewed before completion. When you make a commit, Git takes changes that are in the staging area and make them as a new commit. We are allowed to add and remove changes from the staging area. The staging area can be considered as a place where Git stores the changes. Although, Git doesn't have a dedicated staging directory where it can store some objects representing file changes (blobs). Instead of this, it uses a file called index. Another feature of Git that makes it apart from other SCM tools is that it is possible to quickly stage some of our files and commit them without committing other modified files in our working directory.
- **Maintain the clean history**: Git facilitates with Git Rebase; It is one of the most helpful features of Git. It fetches the latest commits from the master branch and puts our code on top of that. Thus, it maintains a clean history of the project.

Benefits of Git: A version control application allows us to keep track of all the changes that we make in the files of our project. Every time we make changes in files of an existing project, we can push those changes to a repository. Other developers are allowed to pull your changes from the repository and continue to work with the updates that you added to the project files. Some significant benefits of using Git are as follows:

- **Saves Time**: Git is lightning fast technology. Each command takes only a few seconds to execute so we can save a lot of time as compared to login to a GitHub account and find out its features.
- **Offline Working**: One of the most important benefits of Git is that it supports offline working. If we are facing internet connectivity issues, it will not affect our work. In Git, we can do almost everything locally. Comparatively, other CVS like SVN is limited and prefer the connection with the central repository.
- **Undo Mistakes**: One additional benefit of Git is we can Undo mistakes. Sometimes the undo can be a savior option for us. Git provides the undo option for almost everything.
- **Track the Changes**: Git facilitates with some exciting features such as Diff**,** Log**,** and Status, which allows us to track changes so we can check the status**,** compare our files or branches.

Why Git? We have discussed many **features** and **benefits** of Git that demonstrate the undoubtedly Git as the **leading version control system**. Now, we will discuss some other points about why should we choose Git.

- **Git Integrity**: Git is developed to ensure the security and integrity of content being version controlled. It uses checksum during transit or tampering with the file system to confirm that information is not lost. Internally it creates a checksum value from the contents of the file and then verifies it when transmitting or storing data.
- **Trendy Version Control System**: Git is the most widely used version control system. It has maximum projects among all the version control systems. Due to its amazing workflow and features, it is a preferred choice of developers.
- **Everything is Local**: Almost All operations of Git can be performed locally; this is a significant reason for the use of Git. We will not have to ensure internet connectivity.
- **Collaborate to Public Projects**: There are many public projects available on the GitHub. We can collaborate on those projects and show our creativity to the world. Many developers are collaborating on public projects. The collaboration allows us to stand with experienced developers and learn a lot from them; thus, it takes our programming skills to the next level.
- **Impress Recruiters**: We can impress recruiters by mentioning the Git and GitHub on our resume. Send your GitHub profile link to the HR of the organization you want to join. Show your skills and influence them through your work. It increases the chances of getting hired.

GitHub: GitHub is a Git repository hosting service. GitHub also facilitates with many of its features, such as access control and collaboration. It provides a Web-based graphical interface. GitHub is an American company. It hosts source code of your project in the form of different programming languages and keeps track of the various changes made by programmers. It offers both **distributed version control and source code management (SCM)** functionality of Git. It also facilitates with some

collaboration features such as bug tracking, feature requests, task management for every project.

Features of GitHub: GitHub is a place where programmers and designers work together. They collaborate, contribute, and fix bugs together. It hosts plenty of open source projects and codes of various programming languages. Some of its significant features are as follows.

- Collaboration
- Integrated issue and bug tracking
- Graphical representation of branches
- Git repositories hosting
- Project management
- Team management
- Code hosting
- Track and assign tasks
- Conversations
- Wikisc

Benefits of GitHub: GitHub can be separated as the Git and the Hub. GitHub service includes access controls as well as collaboration features like task management, repository hosting, and team management. The key benefits of GitHub are as follows.

- It is easy to contribute to open source projects via GitHub.
- It helps to create an excellent document.
- You can attract recruiter by showing off your work. If you have a profile on GitHub, you will have a higher chance of being recruited.
- It allows your work to get out there in front of the public.
- You can track changes in your code across versions.

Git vs GitHub:

Git is an open-source distributed version control system which is available for everyone at zero cost. It is designed to handle minor to major projects with speed and efficiency. It is developed to co-ordinate the work among programmers. The version control allows you to track and work together with your team member at the same workspace.

While **GitHub is a Git repository hosting service**. It is a web-based service. GitHub facilitates with all of the features of distributed version control and source code management (SCM) functionality of Git. It also supports some of its characteristics in a single software tool.

To better understand the similarities and differences between Git and GitHub, look at the following points.

Git	GitHub
Git is a distributed version control tool that can manage a programmer's source code history.	GitHub is a cloud-based tool developed around the Git tool.
A developer installs Git tool locally.	GitHub is an online service to store code and push from the computer running the Git tool.
Git focused on version control and code sharing.	GitHub focused on centralized source code hosting.
It is a command-line tool.	It is administered through the web.
It facilitates with a desktop interface called Git Gui.	It also facilitates with a desktop interface called GitHub Gui.
Git does not provide any user management feature.	GitHub has a built-in user management feature.
It has minimal tool configuration feature.	It has a market place for tool configuration.

Git Version Control System: A version control system is a software that tracks changes to a file or set of files over time so that you can recall specific versions later. It also allows you to work together with other programmers.

The version control system is a collection of software tools that help a team to manage changes in a source code. It uses a special kind of database to keep track of every modification to the code.

Developers can compare earlier versions of the code with an older version to fix the mistakes.

Benefits of the Version Control System: The Version Control System is very helpful and beneficial in software development; developing software without using version control is unsafe. It provides backups for uncertainty. Version control systems offer a speedy interface to developers. It also allows software teams to preserve efficiency and agility according to the team scales to include more developers. Some key benefits of having a version control system are as follows.

- Complete change history of the file
- Simultaneously working
- Branching and merging
- Traceability

Types of Version Control System

- Localized version Control System
- Centralized version control systems
- Distributed version control systems